

1. Define **Hamming distance**. Prove that it satisfies all properties of a metric.
2. Explain **Binary Symmetric Channel (BSC)** with neat diagram and examples.
3. Explain **Maximum Likelihood Decoding**. Decode a received word over BSC.
4. Define **finite field**. Construct F_4 using $\mathbb{Z}_2[x]/(1 + x + x^2)$.
5. Construct **addition and multiplication tables** for $\mathbb{Z}_2[x]/(1 + x + x^2)$.
6. Define **linear block code**. Explain encoding and decoding process.
7. Explain **nearest neighbour decoding rule** with solved example.
8. Find **cyclotomic cosets** modulo n with suitable example.
9. Explain **error detection and correction capability** of a code.
10. Prove that the **minimum distance of a code determines its error correcting power**.

or space over finite field
space over vector space
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20. Find all cyclotomic cosets of 3 modulo 11.
21. Explain cyclotomic cosets and their importance in coding theory.

Maximum Likelihood Decoding Rule – exp

Maximum Likelihood Decoding Rule – exp
e, decode the received

Maximum Likelihood decoding

Maximum Likelihood decoding

Maximum Likelihood decoding (theory)

1. Define communication channel. Explain its types.
2. Define Hamming distance. Prove that it is a metric.
 - Prove:
 - $0 \leq d(x, y) \leq n$
 - $d(x, y) = 0 \iff x = y$
 - Symmetry property
 - Triangle inequality
3. Calculate Hamming distance between two binary words.
4. Explain Binary Symmetric Channel (BSC) with diagram.
5. Maximum Likelihood Decoding Rule – explain with example.

Decode received word using **Maximum Likelihood**

Decode received word using **Nearest Neighbour**

Find **Hamming distance** between two given words

Find inverse in Z_p .

Check **vector space validity** over finite field.

1. Define communication channel.
2. What is the characteristic of a field?
3. Define finite field.
4. Define Hamming distance.
5. What is a codeword?
6. Define Binary Symmetric Channel.
7. What is error probability?
8. Define vector space.
9. What is a subspace?
10. Define linear code.

1. Define **communication channel** and its types.
2. Define **finite field** and give examples.
3. Define **vector space over F_q** with properties.
4. Define **subspace**. Prove any subset is a subspace.
5. Define **linearly independent and dependent sets**.
6. Define **Hamming distance**. Find distance between two given codewords.
7. Explain **Binary Symmetric Channel** mathematically.
8. Decode a given word using **nearest neighbour decoding**.
9. Find **inverse of an element** in $\mathbb{Z}_p$.
10. Explain **cyclotomic cosets**.

Construct addition and multiplication tables

$$\mathbb{Z}_2$$

in irreducible polynomials over $\mathbb{Z}_2$.

in construction of finite field using

field. State characteristics
a finite field? Explain with
inverse of an element in $\mathbb{Z}_p$
whether a given set is a vec

1. Find Hamming distance between given binary words.
2. Decode a received word using **Maximum Likelihood Decoding**.
3. Decode a received word using **Nearest Neighbour Rule**.
4. Find **inverse of an element** in $\mathbb{Z}_n$.
5. Construct finite field using polynomial modulo.
6. Check whether a given set is a **vector space over F_3** .
7. Find cyclotomic cosets modulo a given number.